

**BulkCell-GNN: A Bipartite Graph Neural Network for
Cross-Modal Phenotype Transfer between Bulk RNA-Seq
and Single-Cell Data in Colorectal Cancer**

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Objective

The objective of this project is to design and implement **BulkCell-GNN**, a heterogeneous bipartite graph neural network that jointly models bulk RNA-seq samples and single-cell transcriptomic data to enable cross-modal information transfer and interpretable phenotype prediction.

The specific objectives are:

- Construct a bipartite graph representing bulk tumor samples and single cells as distinct node types connected through similarity-based edges
- Design a cross-modal message passing mechanism to transfer phenotype information from bulk samples to cell-level representations
- Develop a cell-type-aware aggregation mechanism to explicitly capture the contribution of different cell populations to bulk tumor representations
- Train and evaluate the model for MSI/MSS classification with interpretable outputs aligned with colorectal cancer tumor microenvironment biology

Introduction

Colorectal cancer (CRC) exhibits two clinically distinct molecular subtypes, microsatellite instable (MSI) and microsatellite stable (MSS), which differ substantially in prognosis, immune activity, and response to immunotherapy. Two complementary transcriptomic modalities capture this heterogeneity. Bulk RNA-seq (GSE39582, n=566) provides reliable clinical phenotype labels but averages expression across millions of cells, masking the cellular composition that drives clinical outcomes. Single-cell RNA-seq (GSE132465, n=63,689 cells, 23 patients) resolves individual cell types in the tumor microenvironment at high resolution but typically lacks patient-level phenotype annotations.

Recent computational approaches have explored integrating these modalities through transfer learning, embedding alignment, or joint modeling strategies. However, many methods treat one modality as primary and the other as auxiliary. Bulk-based models achieve strong predictive performance but cannot explain which cell populations drive predictions, whereas single-cell approaches capture cellular heterogeneity but cannot leverage clinical phenotype labels for supervision.

These limitations highlight the need for frameworks that enable direct, bidirectional information flow between bulk-level phenotypes and single-cell representations within a unified model. BulkCell-GNN addresses this need by jointly modeling both modalities as co-equal nodes in a heterogeneous graph and introducing a cell-type-aware aggregation mechanism that enables interpretable linking of cellular composition to bulk tumor phenotypes.

Background

Graph neural networks (GNNs) have become a powerful framework for transcriptomic data analysis, particularly for modeling relationships among genes or cells. scGNN (Wang et al.,

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2021) introduced a graph-based approach for single-cell RNA-seq analysis that uses multi-modal autoencoders and cell-cell graphs, while scGNN 2.0 (Wang et al., 2022) improved performance with attention-based architectures and optionally incorporated bulk data as a deconvolution constraint. However, bulk samples are not explicitly represented as graph nodes, and bidirectional cross-modal interactions are not modeled.

Other approaches, such as scBiG (Luo et al., 2024) and scBGEDA (2023), employ bipartite graph representations for single-cell data, while ATSDP-NET (Zhou et al., 2025) integrates bulk and single-cell data via transfer learning for drug response prediction. Multi-omics GNN frameworks, such as scapGNN (Han et al., 2023) and GNNRAI (2025), further demonstrate the potential of graph-based integration but primarily operate within a single modality or treat modalities asymmetrically.

Collectively, these approaches highlight the growing interest in graph-based and cross-modal modeling, yet they do not explicitly represent bulk samples and single cells as co-equal nodes in a unified heterogeneous graph with bidirectional, cell-type-aware interactions. This motivates the design of BulkCell-GNN, which explores this representation.

Methodology

A heterogeneous bipartite graph is constructed to jointly represent bulk RNA-seq samples and single cells from publicly available colorectal cancer datasets (GSE39582 and GSE132465). The graph comprises two node types — bulk samples and single cells — linked by edges that capture both intra-modality similarity and cross-modal relationships.

The overall workflow consists of three stages: graph construction, cross-modal encoding, and structured message passing.

Graph Construction

Intra-bulk edges connect samples based on similarity in pathway-level expression profiles, capturing patient-level transcriptional structure. Intra-cell edges are constructed using a k-nearest-neighbor graph on reduced-dimensional single-cell expression, modeling local cellular relationships. Cross-modal edges link each bulk sample to transcriptionally similar cells based on shared gene expression, forming the core bipartite structure that enables interaction between modalities.

Representation Learning and Message Passing

Both bulk and single-cell data are encoded into a shared latent space via modality-specific neural encoders. The model then applies a multi-layer graph neural network with three complementary message-passing mechanisms:

- **Intra-bulk aggregation** using graph attention to capture relationships between similar tumor samples
- **Cross-modal attention (bulk → cell)** to propagate phenotype information from bulk samples to cell-level representations

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- **Cell-to-bulk aggregation**, integrating cellular information back into bulk representations

Cell-Type-Aware Aggregation Mechanism

In the cell-to-bulk aggregation step, cells are grouped by their annotated type (e.g., T cells, macrophages, epithelial cells). Representations within each group are pooled and weighted with learnable attention coefficients. This allows each bulk sample to adaptively emphasize the most relevant cellular components, yielding a biologically interpretable representation.

Training Objective

The model is trained using a multi-task objective combining:

- Classification of MSI/MSS labels
- Reconstruction of cell-level expression
- Alignment between bulk and cell representations

This encourages the model to learn biologically meaningful embeddings while preserving consistency across modalities.

Evaluation

Evaluation focuses on both predictive performance and interpretability. Classification performance is assessed using AUC-ROC for MSI/MSS prediction. Interpretability is evaluated using learned cell-type attention weights (γ_i), which indicate the contributions of different cell populations to bulk tumor phenotypes.

References

- [1] Wang, J., Ma, A., Chang, Y., et al. (2021). scGNN: A novel graph neural network framework for single-cell RNA-seq analysis. *Nature Communications*, 12, 1882. <https://doi.org/10.1038/s41467-021-22197-x>
- [2] Wang, J., et al. (2022). scGNN 2.0: A graph neural network tool for imputation and clustering of single-cell RNA-seq data. *Bioinformatics*, 38(23), 5322–5331. <https://doi.org/10.1093/bioinformatics/btac683>
- [3] Lee, H.O., et al. (2020). Lineage-dependent gene expression programs influence the immune landscape of colorectal cancer. *Nature Genetics*, 52, 594–603. <https://doi.org/10.1038/s41588-020-0636-z>
- [4] Marisa, L., et al. (2013). Gene expression classification of colon cancer into molecular subtypes. *PLOS Medicine*, 10(5), e1001453. <https://doi.org/10.1371/journal.pmed.1001453>
- [5] Luo, X., et al. (2024). scBiG: Representation learning of single-cell gene expression data based on bipartite graph embedding. *Briefings in Bioinformatics*, 25(2). <https://doi.org/10.1093/bib/bbae023>

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- [6] Zhou, et al. (2025). ATSDP-NET: Attention-based transfer learning for enhanced single-cell drug response prediction. *Frontiers in Medicine*, 12. <https://doi.org/10.3389/fmed.2025.1631898>
- [7] Han, X., et al. (2023). scapGNN: A GNN-based framework for active pathway and gene module inference from single-cell multi-omics data. *PLOS Biology*, 21(11), e3002369. <https://doi.org/10.1371/journal.pbio.3002369>
- [8] Veličković, P., et al. (2018). Graph Attention Networks. *International Conference on Learning Representations (ICLR)*. <https://arxiv.org/abs/1710.10903>
- [9] Ektefaie, Y., et al. (2023). Multimodal learning with graphs. *Nature Machine Intelligence*, 5, 340–350. <https://doi.org/10.1038/s42256-023-00624-6>
- [10] Yang, H., et al. (2025). Multimodal deep learning approaches for precision oncology. *Briefings in Bioinformatics*, 26(1), bbae699. <https://doi.org/10.1093/bib/bbae699>